

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

Chapter 12: Surface Areas and Volumes

Previous year board questions, topic-wise: 3 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Surface area

1. A solid sphere is cut into two hemispheres. Find the ratio of the surface area of the sphere to the total surface area of the two hemispheres together. (a) 1 : 1 (b) 1 : 4 (c) 2 : 3 (d) 3 : 2

CBSE 2024, 1 mark

2 Volume

2. The largest possible right circular cone is cut from a solid cube of edge 2 cm. Find its volume. (a) $\frac{4\pi}{3}$ cu cm (b) $\frac{5\pi}{3}$ cu cm (c) $\frac{8\pi}{3}$ cu cm (d) $\frac{2\pi}{3}$ cu cm CBSE 2024, 1 mark
3. A hollow cylinder is 14 cm long. Its outer radius is 1 cm more than its inner radius, and the metal in it has volume 176 cm³. Find the two radii. CBSE 2024, 3 marks

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